HEPATIC EPITHELIOID HEMANGIOENDOTHELIOMA — CLINICAL PROTOCOL BOUNDARY ADDENDUM V7

Companion to the existing standard/guidance-aligned protocol, not a relabeling as high risk. The original clinical protocol is unchanged. These OPEN boundary recommendations require ratification before being incorporated into clinical use.

Version 7 | Revised 2026-10-24 06:55 CDT | Institutional approval pending

POLICY AND ALLOCATION STATUS

Analyst A structured critical review through 24 October 2026. All 27 B decisions remain OPEN. Primary evidence, selected-recipient outcomes, pathway-level outcomes, guideline wording and local safeguards are distinguished. This is not a PRISMA-complete systematic review, meta-analysis, fresh external-analyst consensus or institutional approval.

Policy posting checked 24 October 2026: OPTN effective 1 October 2026; oncology guidance December 2025. Clinical suitability, allocation access, donor protection and research authorization remain separate. Recheck before any application. [1] [2]

Use the adjacent study tables before interpreting a proposed cutoff. Abstract-only extraction and missing full PDFs are labeled; missing outcome estimates are not inferred from graphs or from one minus disease-free survival. Linked full text is in this document's own PDF depot when available.

HEHE-B1 — WHEN IS OBSERVATION MORE INFORMATIVE THAN IMMEDIATE LISTING?

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QUESTION AND COMPARISON

For clinically stable, unresectable hepatic EHE, does immediate LT improve survival over observation-informed selection, and is a four-month delay evidence-based?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Lai 2017 [3] Retrospective ELTR/ELITA; 149 adult LT recipients, 1984–2014; median post-LT follow-up 7.6 years. Primary source | Full-text PDF Locator: Results, Table 5 and Figure 3; use post-LT, not diagnosis-based OS. Review depth: Primary full text reviewed; local complete PDF available	**Reported result:** Post-LT 5-year OS 79.5%; 37/149 recurred. Waiting ≤120 days associated with recurrence: Table 5 adjusted HR 2.5 (95% CI 1.2–5.3); 5-year DFS 73.3% versus 86.7% with longer waiting. **Do not infer:** Confounding by urgency and survivorship; association does not show that waiting causes benefit. Earlier ELTR series overlap. Abstract HR differs slightly from Table 5.

Study • population • source	Results • interpretation boundary
Brahmbhatt 2020 [4] UNOS 2002–2018: 131 adults listed; 88 transplanted. Primary source | TMC: retrieve PDF Locator: Published abstract Results; PMID 31 577 674. Review depth: Primary published abstract verified; complete PDF pending	**Reported result:** Post-LT 1/3/5-year OS 88.6%/78.9%/77.2%. Early hepatic-artery-thrombosis graft failure 4.6% versus 0.5% in non-HEHE comparators. **Do not infer:** Abstract-only extraction. Favorable survival despite median wait <120 days argues against a proven mandatory four-month threshold; not a randomized timing comparison. Older UNOS reports overlap.
Thomas 2014 [5] Single-center retrospective cohort, n=50; initial surgery 7, systemic/embolization 18, observation 25. Primary source | TMC: retrieve PDF Locator: Methods time-origin definition; Table 1 and survival Results. Review depth: Primary full HTML Methods, Results and tables reviewed; complete PDF pending	**Reported result:** Five-year OS from diagnosis: surgery 83.3%, systemic/embolization 71.3%, observation 72.4% (P=.9). Localized versus metastatic: 69% versus 78.3% (P=.7). **Do not infer:** Small, nonrandom groups; no metastatic patients in initial surgery group. Supports observing selected indolent disease, not equivalence of therapies or a 120-day cutoff.
Liu 2024 [6] Single-team retrospective cohort 2014–2024: 228 evaluable; LT 8, resection/ablation 49, nonsurgical 171. Primary source | Full-text PDF Locator: Methods; Tables 1–3; Survival Outcomes/Figure 2. Review depth: Primary full HTML and complete publisher PDF reviewed; Methods and Tables 1–5	**Reported result:** Overall 5-year OS 84.9%; LT 1/3-year OS 62.5%/25%; non-LT 5-year OS 87.8%. Survival time zero is not clearly specified in the extracted Methods; do not relabel as standard post-LT estimates. **Do not infer:** Major confounding by indication, eight recipients and loss-to-follow-up exclusion. Poor LT results cannot prove LT causes harm; they challenge assuming all HEHE has excellent transplant outcomes.
Stacchiotti 2021 [7] International expert consensus; evidence predominantly retrospective and very-low certainty. Primary source | Full-text PDF Locator: Management of liver EHE, PDF p8; active-surveillance discussion. Review depth: Primary full text reviewed; local complete PDF available	**Reported result:** Supports initial active surveillance for suitable stable disease and consideration of LT for selected unresectable liver-only EHE; advises against LT with EHD (V,A). **Do not infer:** Expert recommendation conflicts with ACG key concepts and NLRB guidance. Neither side has randomized EHD transplant-benefit evidence.

Study • population • source	Results • interpretation boundary
DGVS S2k LT guidance [8] German S2k consensus guideline; HEHE recommendations dated 2023, publication 2024. Primary source | TMC: retrieve PDF Locator: Official society HEHE chapter, recommendations 3.85–3.86. Review depth: Official society HTML HEHE chapter reviewed; consensus recommendations dated 2023; complete guideline PDF pending	**Reported result:** Recommends at least six months of largely stable disease before LT. Macrovascular invasion or hilar nodes are not absolute barriers to evaluation, which is not equivalent to recommending transplantation. **Do not infer:** Expert consensus, not a randomized timing comparison or US allocation rule. It supplies a competing six-month option, not proof of superiority to four months.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

Observation can expose indolent biology and avoid unnecessary graft use. The 120-day signal is prognostic association among recipients, not a tested intervention. Good US outcomes with a shorter median wait and poor outcomes in a small high-burden cohort both argue for clinical context rather than a stopwatch. DGVS explicitly favors six months of largely stable disease. This expert-guidance counterpoint is more conservative than the earlier four-month local proposal; neither interval has causal comparative validation. [3] [4] [5] [6] [7] [8]

- There is no randomized immediate-versus-delayed LT comparison. Time accrued before referral, urgent liver failure, selection to survive waiting, lead time and different OS clocks can generate apparent benefits of observation.

Certainty / applicability: Low certainty for selected post-LT survival; very low certainty that a fixed observation interval improves outcomes. No validated universal 120-day eligibility cliff. Lai measured waiting time from waiting-list registration, not completed staging, diagnosis or referral; our proposed staging-based clock is an explicit adaptation.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- Six months: international expert-guidance default, not US policy or a proven biological cliff. Reliable pre-referral stability may contribute.
- 120 days: Lai registry exposure measured from waiting-list registration to LT; it is not the same as diagnosis-to-treatment or completed-staging observation and does not causally validate four months.
- Approximately eight-week reassessment and exceptions: local workflow/safety judgment; adjust to disease trajectory.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
Lai 2017 [3] “pre-LT waiting time of 120 days or less” Published PDF p1, abstract Results Source | Full-text PDF	Associated with recurrence in a selected transplant registry; the association does not prove that delaying LT improves outcomes.

Reference and exact excerpt	What it supports / does not establish
DGVS S2k LT guidance [8] “mindestens 6 Monaten einen weitgehend stabilen Krankheitsverlauf” Recommendation 3.86, official society HEHE chapter Source | Local PDF pending	Short original German excerpt: the guidance calls for a largely stable course for at least six months. This is expert consensus, not US allocation policy.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Use six months of documented stability as an OPEN default for clinically stable HEHE, with explicit exceptions and continued assessment of whether LT is needed. [3] [8] [4] [5] [6] [7]

- In stable disease, propose six months of largely stable biology before final LT activation, with comparable staging and approximately eight-week reassessment.
  - Reliable earlier imaging can count; document the actual diagnosis, staging, registration and observation dates rather than restart a clock at referral.
- Do not delay threatening liver failure, significant progressive symptoms or rapid progression solely to complete six months.
  - Obtain urgent multidisciplinary review of LT versus non-LT treatment; urgency does not itself establish oncologic suitability.
- Six months is an expert-guidance-informed local default, not a national requirement or validated cutoff.
  - Continued stability does not mandate LT: reassess resection, symptom burden, function and benefit over continued observation.

Why retain or change it: Revise the OPEN default to six months of documented stability in clinically stable patients, informed by DGVS consensus. Count reliable pre-referral comparable imaging rather than restart an arbitrary clock. Reassess at roughly eight-week intervals; threatening disease requires urgent individualized review, not automatic delay or automatic LT. Continued indolence can still favor observation.

INFORMATION REQUIRED BEFORE THE DECISION

Expert EHE pathology; complete liver/systemic staging; symptoms and liver function; interval growth and vascular/node assessment; resection review.

Clinical versus allocation: This is a proposed clinical timing choice, not an added NLRB exception requirement. Standard HEHE protocol content is not rewritten in this edition.

Roadmap / protocol reconciliation: Use the existing HEHE standard pathway and specialty review; this meeting record is included to answer every boundary question.

Meeting record: HEHE-B1 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

HEHE-B2 — CAN EXTRAHEPATIC DISEASE OR AN EXPLANT SCORE BROADEN ACCESS?

HEHE-B2 — CAN EXTRAHEPATIC DISEASE OR AN EXPLANT SCORE BROADEN ACCESS?

QUESTION AND COMPARISON

For unresectable EHE with EHD or vascular/nodal risk, can selected outcomes or an explant score justify broader pre-LT access?

EVIDENCE BEFORE CHOOSING A RULE

Study • population • source	Results • interpretation boundary
Lai 2017 [3] Same 149-recipient ELTR series; 40 had pre-LT extrahepatic disease. Primary source | Full-text PDF Locator: Table 5; propensity-matched analysis; EHE-LT score section. Review depth: Primary full text reviewed; local complete PDF available	**Reported result:** Matched 5-year DFS: EHD 73.5% versus 81.6% without, P=.54. macrovascular invasion adjusted recurrence HR 4.9 (2.4–9.9); hilar nodes HR 2.2 (1.1–4.5). Score groups 0–2/3–5/6–10: 5-year DFS 93.9%/76.9%/38.5%. **Do not infer:** A nonsignificant small comparison is not equivalence. Score was not validated as a pre-LT eligibility test; grade/stage can be missed before explant.
ACG 2024 [9] ACG 2024 focal liver lesion guideline, key concepts 34–36. Primary source | Full-text PDF Locator: Table 3 key concepts 34–36, PDF p5; EHE discussion. Review depth: Complete local PDF reviewed; HCA/EHE key concepts and supporting text; key concepts are not GRADE-rated recommendations	**Reported result:** Key concept 35 supports resection if possible and LT when unresectable, including selected EHD. This is a clinical position, not an allocation entitlement. **Do not infer:** These statements are key concepts, not GRADE-rated recommendations; optimistic comparative language rests on selected series.
Stacchiotti 2021 [7] International expert consensus; evidence predominantly retrospective and very-low certainty. Primary source | Full-text PDF Locator: Management of liver EHE, PDF p8; active-surveillance discussion. Review depth: Primary full text reviewed; local complete PDF available	**Reported result:** Supports initial active surveillance for suitable stable disease and consideration of LT for selected unresectable liver-only EHE; advises against LT with EHD (V,A). **Do not infer:** Expert recommendation conflicts with ACG key concepts and NLRB guidance. Neither side has randomized EHD transplant-benefit evidence.

Study • population • source	Results • interpretation boundary
NLRB oncology guidance [2] Official December 2025 transplant-oncology exception guidance, current posting checked 24 October 2026. Primary source | Full-text PDF Locator: HEHE section p11 and disclaimer p2. Review depth: Complete local primary PDF available; relevant methods, selection, results and limitations reviewed	**Reported result:** EHD is not an absolute contraindication in this guidance. No mandatory four-month observation rule is specified. **Do not infer:** Nonbinding allocation-review guidance; does not demonstrate individual clinical benefit or override donor/clinical judgment.
Liu 2024 [6] Single-team retrospective cohort 2014–2024: 228 evaluable; LT 8, resection/ablation 49, nonsurgical 171. Primary source | Full-text PDF Locator: Methods; Tables 1–3; Survival Outcomes/Figure 2. Review depth: Primary full HTML and complete publisher PDF reviewed; Methods and Tables 1–5	**Reported result:** Overall 5-year OS 84.9%; LT 1/3-year OS 62.5%/25%; non-LT 5-year OS 87.8%. Survival time zero is not clearly specified in the extracted Methods; do not relabel as standard post-LT estimates. **Do not infer:** Major confounding by indication, eight recipients and loss-to-follow-up exclusion. Poor LT results cannot prove LT causes harm; they challenge assuming all HEHE has excellent transplant outcomes.
DGVS S2k LT guidance [8] German S2k consensus guideline; HEHE recommendations dated 2023, publication 2024. Primary source | TMC: retrieve PDF Locator: Official society HEHE chapter, recommendations 3.85–3.86. Review depth: Official society HTML HEHE chapter reviewed; consensus recommendations dated 2023; complete guideline PDF pending	**Reported result:** Recommends at least six months of largely stable disease before LT. Macrovascular invasion or hilar nodes are not absolute barriers to evaluation, which is not equivalent to recommending transplantation. **Do not infer:** Expert consensus, not a randomized timing comparison or US allocation rule. It supplies a competing six-month option, not proof of superiority to four months.

SYNTHESIS AND EVIDENCE THAT CHALLENGES THE PROPOSAL

The disagreement is genuine: ACG/NLRB are permissive about selected EHD, whereas the disease-specific consensus is restrictive. The best registry suggests macrovascular invasion and hilar nodes are adverse; selected EHD was not independently adverse, but the study was too small to establish equivalence. Histology-derived risk scoring cannot be substituted for preoperative staging. [3] [9] [7] [2] [6] [8]

- A blanket statement that any EHD absolutely forbids LT is inaccurate. Conversely, favorable survival in selected EHD recipients does not establish benefit over observation or systemic options for all sites/burdens.

Certainty / applicability: Low certainty for prognostic association; very low for causal EHD benefit or pre-LT score-driven selection.

WHERE EACH PROPOSED THRESHOLD COMES FROM

- No macrovascular invasion: aligns with current favorable NLRB consideration and adverse ELTR prognosis.
- Liver-only opening cohort: conservative local choice despite permissive ACG/NLRB wording; cannot be called universally guideline-mandated.
- EHE-LT score cutoffs: post-LT risk stratification, not a validated pre-LT exclusion test.

SOURCE WORDING — READ IN CONTEXT

Reference and exact excerpt	What it supports / does not establish
NLRB oncology guidance [2] "The presence of extrahepatic disease is not an absolute contraindication." December 2025 oncology guidance p11, HEHE Source | Full-text PDF	Allocation guidance permits consideration; it does not establish clinical benefit in every EHD phenotype.
NLRB oncology guidance [2] "Absence of macrovascular invasion on biopsy or imaging." December 2025 oncology guidance p11, HEHE Source | Full-text PDF	A separate favorable-guidance criterion; do not omit it when discussing EHD.
Stacchiotti 2021 [7] "LT should not be proposed to patients with liver EHE and extrahepatic disease" ESMO Open 2021 consensus PDF p8; recommendation graded V, A Source | Full-text PDF	Clinical consensus is more restrictive than NLRB review guidance. The disagreement remains visible.
ACG 2024 [9] "even in the setting of extrahepatic disease" Table 3, key concept 35, PDF p5 Source | Full-text PDF	ACG's key-concept wording is more permissive than the EHE consensus; it is not a graded comparative trial result.

Excerpts are deliberately short; use the linked complete source for context. Thresholds labeled proposed are not quotations or institutional policy.

OPEN RECOMMENDATION AFTER THE EVIDENCE REVIEW

Proposed stance: Start with liver-only disease and no macrovascular invasion; do not automatically broaden access using stable extrahepatic disease or an explant score. [2] [7] [9] [3] [6] [8]

- Proposed opening boundary: unresectable liver-only EHE with no macrovascular invasion.
  - Extrahepatic disease is a separate unactivated stratum requiring case-specific discussion, not a national absolute prohibition.
- Resolve suspicious nodes/extrahepatic lesions before proceeding.

  - Do not use an explant-derived score as if it were a validated pretransplant eligibility test.

- Record the guideline disagreement explicitly.

  - A future EHD proposal must specify site, control duration, symptoms, comparator and independent review; “stable EHD” alone is insufficient.

Why retain or change it: Retain liver-only/no-macrovascular invasion as an OPEN conservative initial local boundary, explicitly not a national absolute prohibition. Consider EHD only through a separately adjudicated proposal with site, behavior and non-LT comparator specified.

INFORMATION REQUIRED BEFORE THE DECISION

Complete staging; expert pathology; vascular assessment; targeted evaluation of suspicious extrahepatic sites; technical resectability and alternative-treatment review.

Clinical versus allocation: Clinical selection and exception-review eligibility answer different questions. The proposed liver-only choice is local, not a rewrite of national guidance.

Roadmap / protocol reconciliation: Preserve the distinction in the standard HEHE alignment record. No automatic clinical expansion is made by this document.

Meeting record: HEHE-B2 | Readable evidence / decision text

OPEN — no committee decision, clinical adoption or transplanting-study activation is implied.

Evidence review record: Five analyst comments and reconciliation | Prospective outcomes dataset

REFERENCES

[1] Organ Procurement and Transplantation Network. OPTN Policies: liver allocation and living donation. HRSA/OPTN. 2026. Official source | Full text PDF

Policy or professional guidance, not a primary treatment study. Official full PDF checked.

[2] Organ Procurement and Transplantation Network. Adult MELD Exceptions for Transplant Oncology. HRSA/OPTN. 2025. Official source | Full text PDF

Allocation review guidance. Complete local primary PDF available; relevant methods, selection, results and limitations reviewed.

[3] Lai, Quirino; Feys, Estelle; Karam, Vincent; et al. Hepatic Epithelioid Hemangioendothelioma and Adult Liver Transplantation: Proposal for a Prognostic Score Based on the Analysis of the ELTR-ELITA Registry. Transplantation. 2017;101(3):555-564. DOI | Full text PDF

Retrospective transplant registry. Primary full text reviewed; local complete PDF available.

[4] Brahmbhatt, Mihir; Prenner, Stacey; Bittermann, Therese. Liver Transplantation for Hepatic Epithelioid Hemangioendothelioma Is Facilitated by Exception Points With Acceptable Long-term Outcomes. Transplantation. 2020;104(6):1187-1192. DOI | Local PDF pending

Clinical research / guidance. Primary published abstract verified; complete PDF pending.

[5] Thomas, Ryan M.; Aloia, Thomas A.; Truty, Mark J.; et al. Treatment sequencing strategy for hepatic epithelioid haemangioendothelioma. HPB. 2014;16(7):677-685. DOI | Local PDF pending

Clinical research / guidance. Primary full HTML Methods, Results and tables reviewed; complete PDF pending.

[6] Liu, Xiaolei; Yang, Peijun; Liu, Liguo; et al. Long-term prognosis and treatment modalities of hepatic epithelioid hemangioendothelioma: a retrospective study of 228 patients. BMC Cancer. 2024;24(1):1285. DOI | Full text PDF

Clinical research / guidance. Primary full HTML and complete publisher PDF reviewed; Methods and Tables 1–5.

[7] Stacchiotti, S.; Miah, A.B.; Frezza, A.M.; et al. Epithelioid hemangioendothelioma, an ultra-rare cancer: a consensus paper from the community of experts. ESMO Open. 2021;6(3):100 170. DOI | Full text PDF

Clinical consensus. Primary full text reviewed; local complete PDF available.

[8] Berg, Thomas; Aehling, Niklas F.; Bruns, Tony; et al. S2k-Leitlinie Lebertransplantation: HEHE recommendations 3.83–3.86. Zeitschrift für Gastroenterologie. 2024;62(09):1397-1573. DOI | Local PDF pending

Clinical research / guidance. Official society HTML HEHE chapter reviewed; consensus recommendations dated 2023; complete guideline PDF pending.

[9] Frenette, Catherine; Mendiratta-Lala, Mishal; Salgia, Reena; et al. ACG Clinical Guideline: Focal Liver Lesions. American Journal of Gastroenterology. 2024;119(7):1235-1271. DOI | Full text PDF

Clinical research / guidance. Complete local PDF reviewed; HCA/EHE key concepts and supporting text; key concepts are not GRADE-rated recommendations.

REVISION NOTE

Revised 2026-10-24 06:55 CDT. Evidence-before-recommendation review of all 27 B questions. Earlier protocol versions and external analyst records are preserved; this is an Analyst A synthesis, not a new external-analyst vote.

Problem: HEHE/HCA B questions required the same evidence depth as the six high-risk protocols.
Correction and reason: Added a blue-format clinical boundary addendum with the full decision-specific evidence, uncertainty and OPEN proposals, without overwriting or relabeling the standard pathway.

Approval record Clinical lead _____ Allocation review _____ Research authorization if required _____ Donor review if applicable _____ Date _____